FinWave

COCOMO Estimation & UML Diagrams

Team 20 | MIT Academy of Engineering | 22 May 2026

Part A: COCOMO Estimation

1.1 What is COCOMO?

COCOMO (Constructive Cost Model) is a software cost estimation model developed by Barry Boehm. It uses Lines of Code (LOC) to estimate effort (in person-months), development time (in months), and team size required to build a software project. For the hackathon, COCOMO helps us quantify our project complexity and planning.

1.2 COCOMO Mode Selection — Organic

FinWave falls under the Organic Mode of COCOMO based on the following criteria:

- Small project team (6 members) with good domain understanding
- Flexible requirements typical of a startup/hackathon environment
- Estimated size between 2,000–3,000 Lines of Code
- Well-understood application domain (fintech web app)
- Relatively relaxed time constraints per module

1.3 LOC Estimation by Module

Module	Component	Estimated LOC	Justification
Authentication	Login, Register, OTP	350	Form handling, JWT logic, validation
Wallet	Dashboard, Balance, Add Funds	300	State management, API calls, UI
Send Money	Transfer form, receipt, validation	400	Core business logic, error handling
Transaction History	List, filter, detail view	250	Data fetching, filtering, display
Language (i18n)	EN/HI/MR translation files + switcher	200	Translation keys, context provider
Financial Tips	Rule engine, tip display	200	Rule conditions, dynamic content
Backend API	Express routes, controllers, middleware	600	RESTful endpoints for all modules
Database Models	User, Wallet, Transaction schemas	250	Mongoose/Firebase schema definitions

Module	Component	Estimated LOC	Justification
Configuration & Misc	Config files, utils, tests	200	Environment setup, helper functions
TOTAL	All Modules	2,750 LOC	

1.4 COCOMO Organic Mode Formulae

The Basic COCOMO Organic Mode equations are:

Effort (E) = $a \times (KLOC)^b$ person-months
Time (T) = $c \times (E)^d$ months
Team Size = E / T people

For Organic Mode, the constants are: a = 2.4, b = 1.05, c = 2.5, d = 0.38

1.5 COCOMO Calculations

Parameter	Value	Formula / Notes
Estimated LOC	2,750	Estimated across all modules
KLOC	2.75	LOC / 1000
Effort (E)	7.11 person-months	2.4 x (2.75)^1.05 = 2.4 x 2.963 = 7.11
Development Time (T)	4.06 months	2.5 x (7.11)^0.38 = 2.5 x 1.624 = 4.06
Recommended Team Size	~2 people	E / T = 7.11 / 4.06 = 1.75 ≈ 2 people
Productivity	386.8 LOC/person-month	KLOC x 1000 / Effort
Actual Team Size	4 people	Hackathon constraint (faster delivery)
Adjusted Time (4 members)	~1.78 months (or ~7 days)	7.11 / 4 = 1.78 person-months per person
Hackathon Duration	1 day (24 hours)	Compressed timeline — MVP scope justified

1.6 COCOMO Justification for Hackathon

The COCOMO estimation validates our project scope. With a full team of 6, the equivalent effort for a 2,750-LOC project is distributed as approximately 1.78 person-months each — confirming that building an MVP with core modules within a 24-hour hackathon is feasible, particularly since:

- We are building a prototype, not production-grade code
- We are using high-productivity frameworks (React, Node.js) that accelerate development
- Module responsibilities are cleanly divided among team members
- Documentation, UML, and non-coding deliverables are equally valued by judges